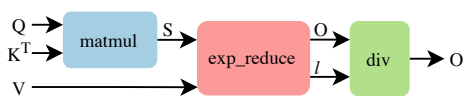


```
for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do
     $S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$ 
```

```
for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do
     $m_i := \max(m_i, S_{i,j})$ 
  for all  $i \in [0, N)$  do
    for  $j \in [0, N)$  do
       $P_{i,j} := \exp(S_{i,j} - m_i)$ 
       $\ell_i := \ell_i + P_{i,j}$ 
       $P_{i,j} := P_{i,j} / \ell_i$ 
```

```
for all  $i, j \in [0, N) \times [0, d)$  do
  for  $k \in [0, N)$  do
     $O_{i,j} := O_{i,j} + P_{i,k}V_{k,j}$ 
```



```
for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do
     $S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$ 
```

```
for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do
```

```
 $\tilde{P} = e^{S_{i,j}}$ 
 $\begin{bmatrix} m_i \\ \ell_i \\ O_i \end{bmatrix} := \begin{bmatrix} m_i \\ \ell_i \\ O_i \end{bmatrix} \oplus_2 \begin{bmatrix} S_{i,j} \\ \tilde{P} \\ \tilde{P}V_j \end{bmatrix}$ 
```

```
for all  $i, j \in [0, N) \times [0, N)$  do
   $O_{i,j} := O_{i,j} / \ell_i$ 
```

decompose

```
for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do
     $S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$ 
```

```
for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do
     $\tilde{m} = \max(m_i, S_{i,j})$ 
     $\tilde{P} = e^{S_{i,j} - \tilde{m}}$ 
     $\tilde{\alpha} = e^{m_i - \tilde{m}}$ 
```

```
 $m_i := \tilde{m}$ 
 $\ell_i := \alpha \ell_i + \tilde{P}$ 
for all  $k \in [0, d)$  do
   $O_{i,k} := \alpha O_{i,k} + \tilde{P}V_{j,k}$ 
```

```
for all  $i, j \in [0, N) \times [0, d)$  do
   $O_{i,j} := O_{i,j} / \ell_i$ 
```